Satellite Simulation Workshop

Introduction

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Aims of this Session

Appreciate the complexity of Computer Systems:

`Software + Hardware + People + Processes`

Focus on the role of human factors in such systems

This will be achieved through:

- The use of purpose-build software simulation
- Investigating a specific historical case study
- Exploring discussion from academic publications

Computer Science

In many complex applications, Computer Science often works in collaboration with other disciplines

Computers are SO useful, everyone is using them !
However this requires knowledge of BOTH disciplines
(Computer Science AND the Application Domain)
Not just two separate people - but people with BOTH

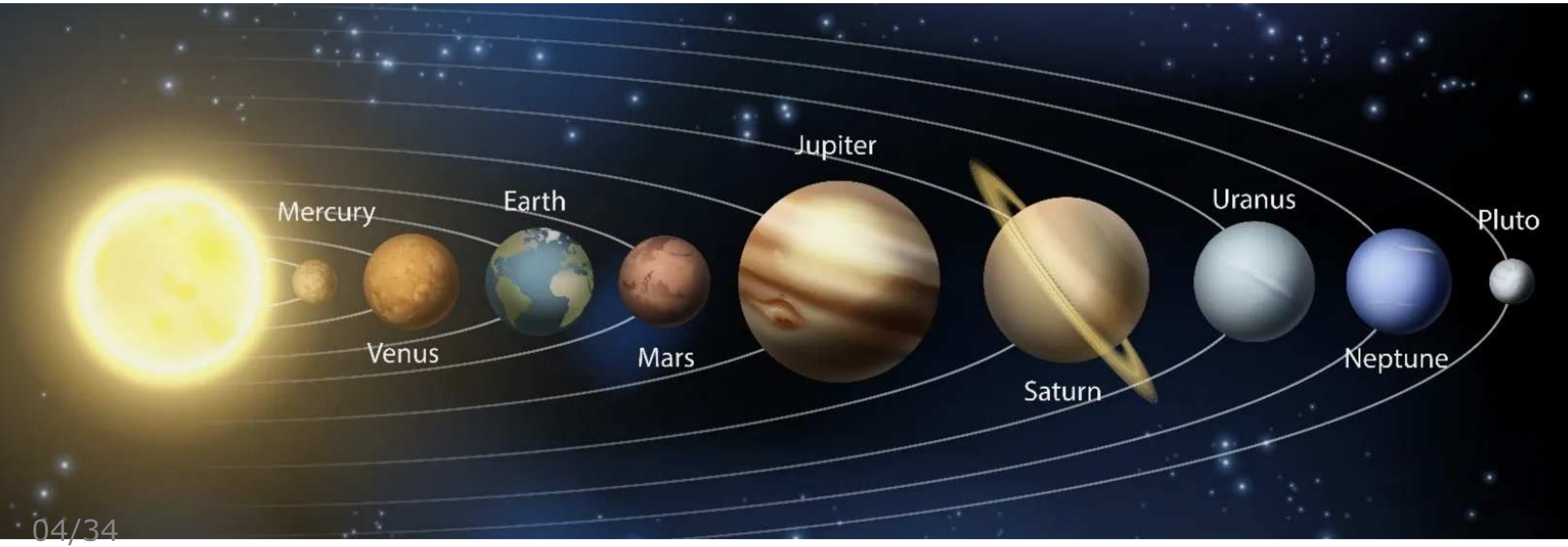
Let's look at one such application domain...

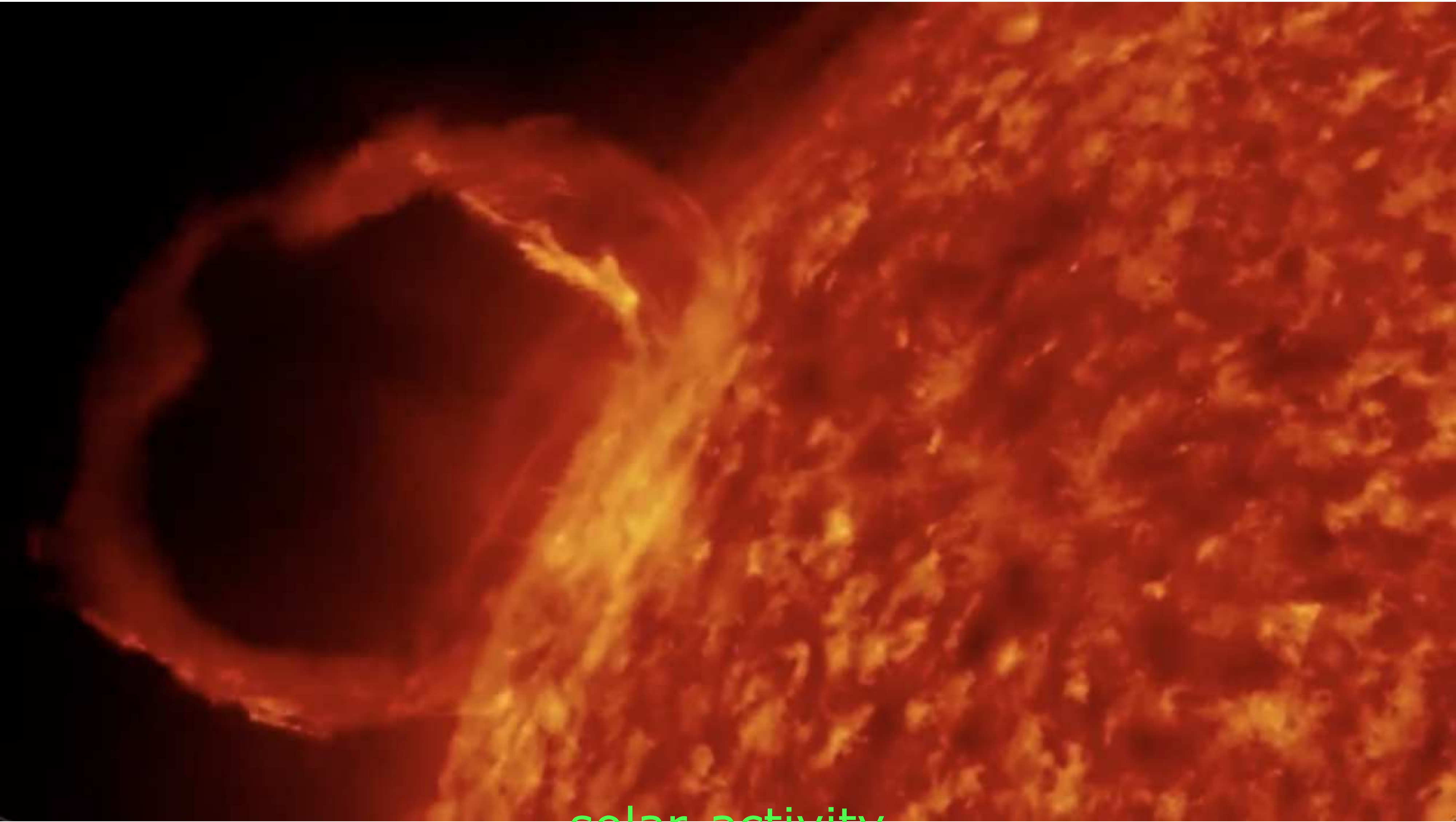
Our Solar System

Let's test your knowledge of our solar system

How many planets are there (8/9/10) ?

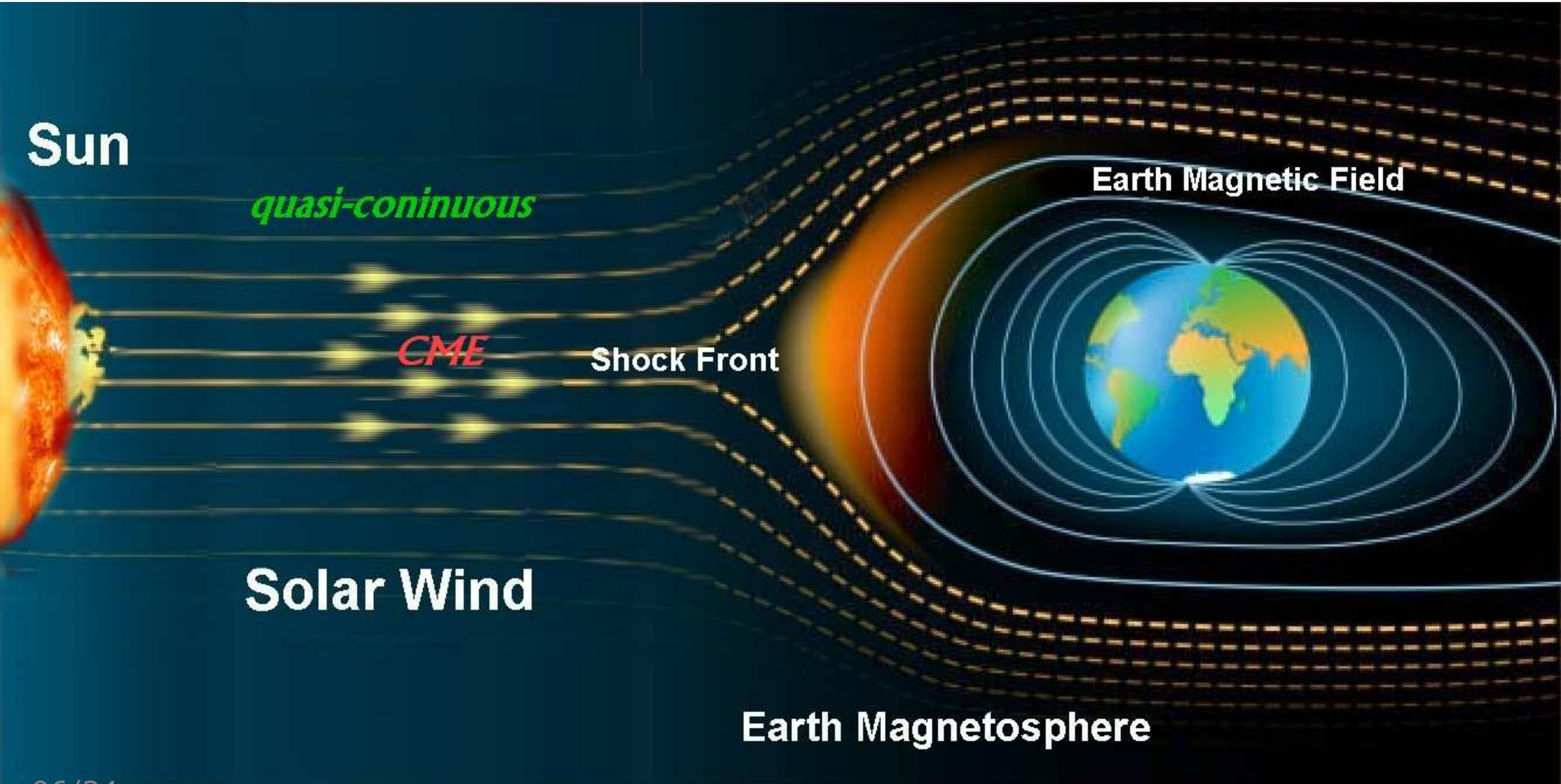
WhichOrbitsWhich





solar-activity

Solar Wind



Sun
quasi-coninuous
CME
Shock Front
Earth Magnetic Field
Solar Wind
Earth Magnetosphere

Aurora Borealis - Northern Lights

Solar Wind

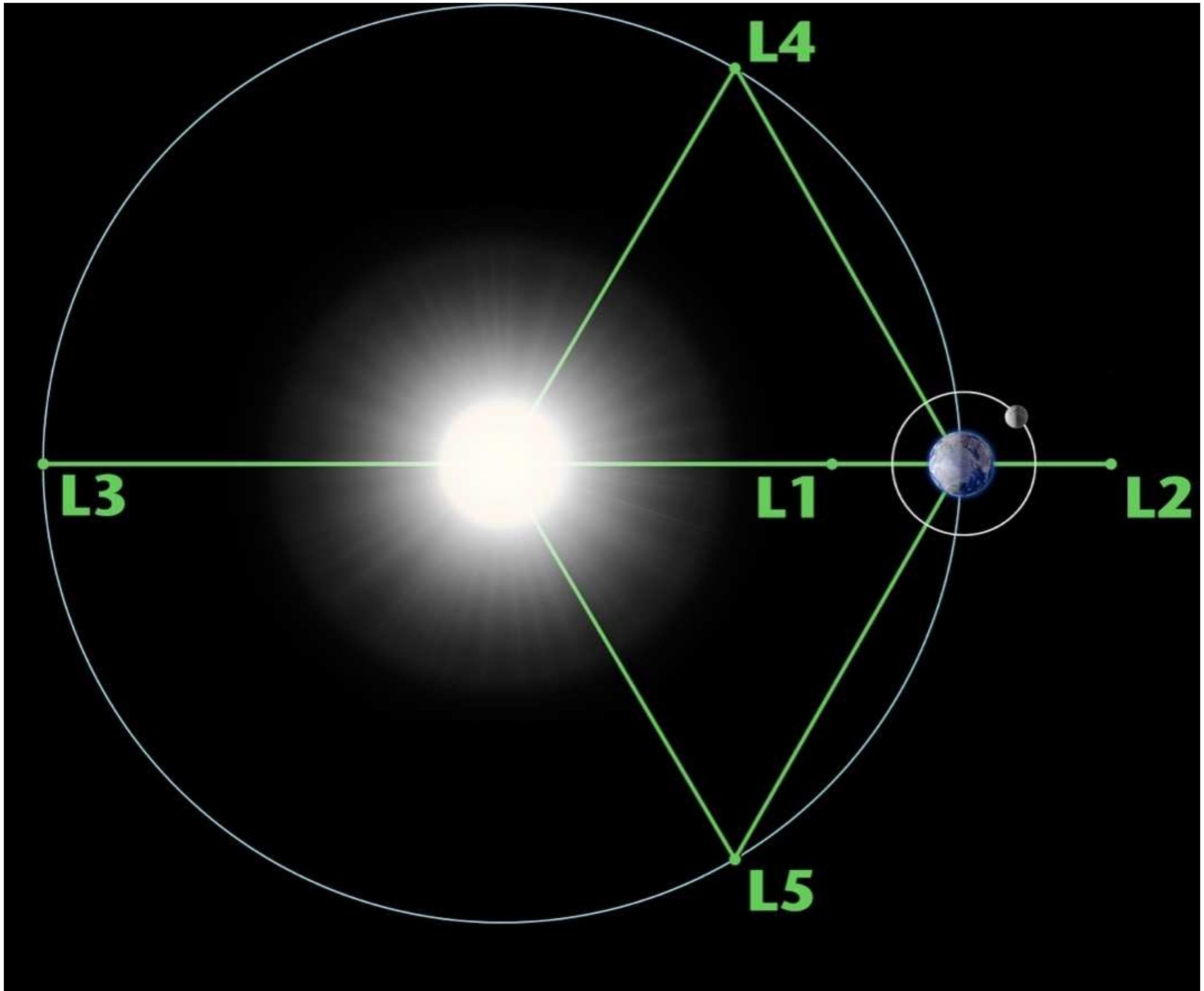
Travels at around 400 km/s (~1.5 million km/h)
(but still takes three or four days to reach Earth)

When it strikes the Earth, solar wind can:

- cause power outages on the ground
- interfere with communication networks
- disrupt operation of satellites
- interfere with navigation systems
- injure astronauts in space !

A good idea to use satellites to monitor !

Lagrange Points (not to scale !)



Lagrange Point "L1"

This is an ideal location for a solar observatory
Good for assessing impact of solar wind on Earth !

L1 is 1.5 million km from the Earth

For reference, Earth is approx.
150 million km from Sun

Takes 4 about months to travel
from Earth to L1 (at 500 km/h)

Satellites currently at L1

Solar and Heliospheric Observatory (SOHO)

Aditya-L1 (arrived in January 2024 !)

Global Geospace Science "Wind" satellite

Deep Space Climate Observatory (DSCOVR)

Advanced Composition Explorer (ACE)...

http://services.swpc.noaa.gov/text/ace-swepam.txt

Graphing

It can be hard to interpret raw numerical data

Instead we can pull the live feed from the satellite

And represent the numerical data visually:

ACEGrapher

Practical Activity

Enough theory and high-level discussion

Let's get hands-on with a practical activity

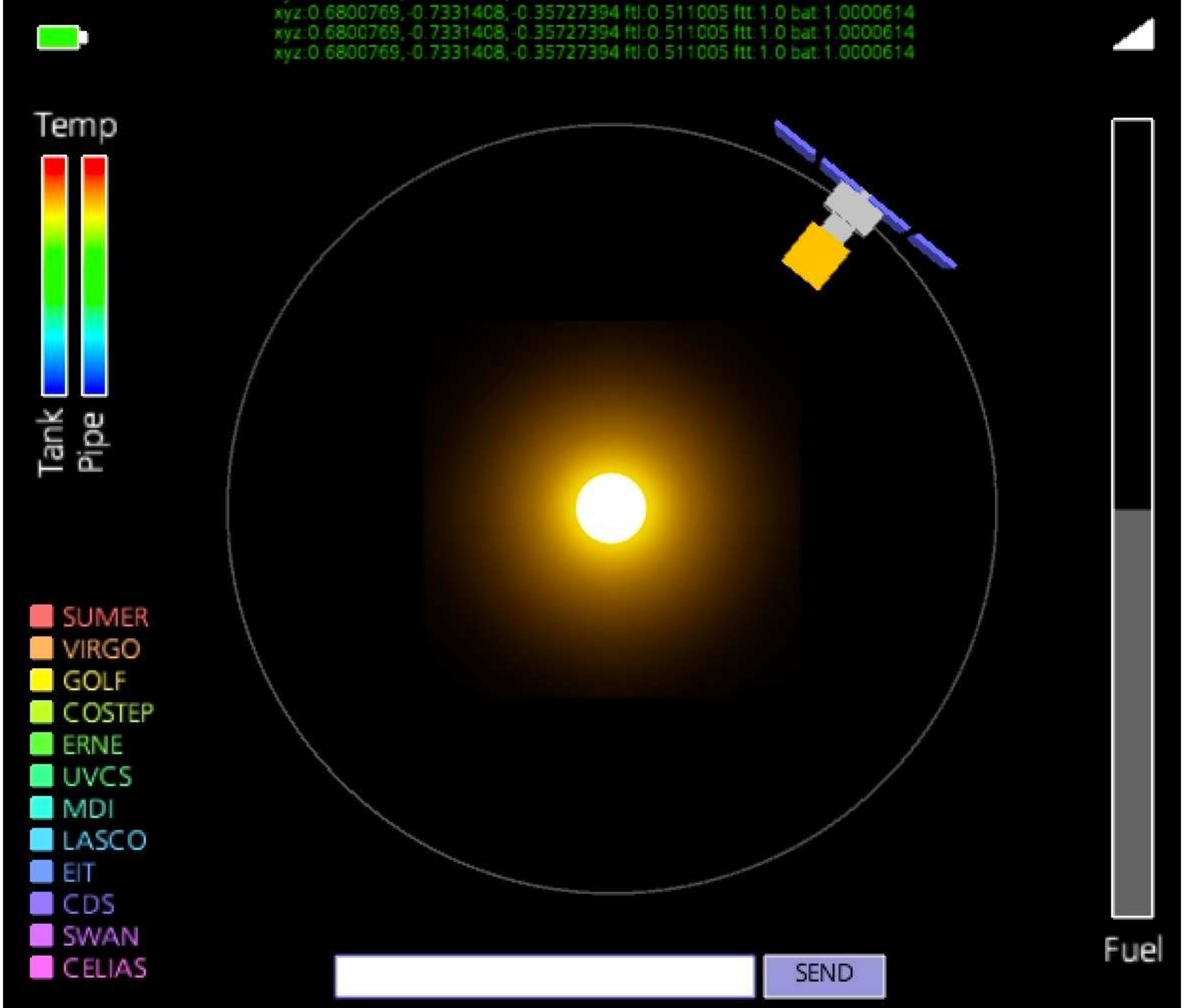
UNFORTUNATELY

We don't have a real satellite to play with :o(

FORTUNATELY

Since we are Computer Scientists...

We can *build* a simulator to experiment with !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Important Note

Note that this simulation is an *approximation*
(not everything is scientifically accurate)

We've made a few compromises to make it usable:

- Scale: can't fit 150 million km on the screen !
- Time: sped up animation to reduce waiting time
- Dimensions: 2D rather than 3D for easier interaction
- Simplified Commands: to make more understandable

Simulator downloaded

Simulator written in Processing

But exported as an application !

https://tinyurl.com/BristolSun

Download it, UNZIP IT, run it:

`SatelliteController`

Will need to manage security

Depending on your platform...

Windows Security

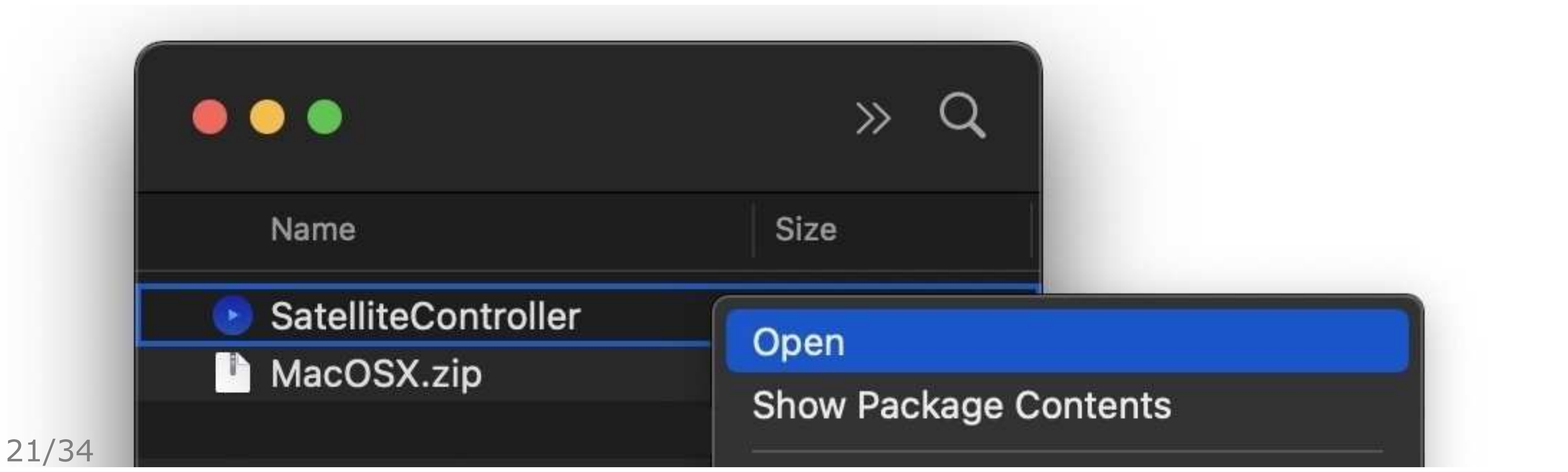
MacOSX Security

Download then Unzip MacOSX in Downloads folder

Important: move SatelliteController to suitable folder

Control-Click on the SatelliteController file

Then select "Open" from the menu



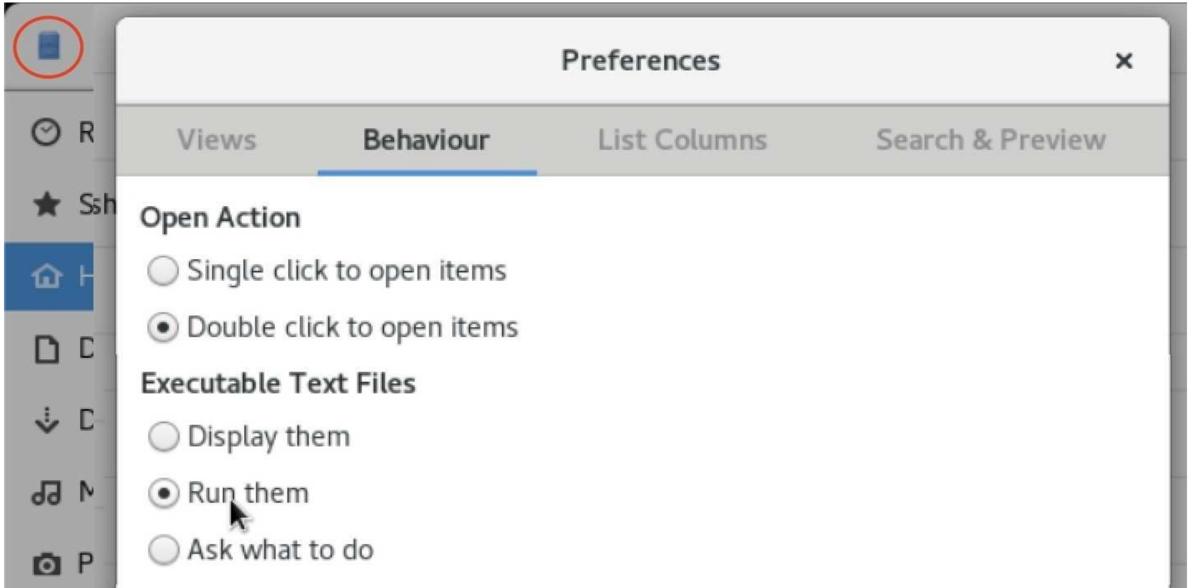
Linux Security (including lab machines)

Open up the terminal and then type "nautilus"

When file explorer opens, go to "preferences" menu

For "Behaviour > Exec Text Files" select "Run them"

You can now run SatelliteSimulator by double-clicking



[Wait for everyone to start simulator]

A range of control commands are available

Let's open up the simulation to try them out:

SatelliteController

Run your simulator and follow along on your computer

As we step through commands on following slides...

Science Experiments

Satellite carries a range of scientific experiments
We won't explore the nature of these experiments
(they are space science, this is computer science)

We switch OFF/ON all science experiments using:

`SCI0`

`SCI1`

Useful if we need to save some power, however...
Scientists won't be very happy if we leave them off

WARNING

We are about to use some navigation commands

It is ESSENTIAL that we are VERY careful with these

We MUST NOT let satellite get out of alignment

Move too much and we will lose solar power

Might lose radio signal (due to misaligned antenna)

BE CAREFUL

Only move satellite by small amounts

Always move back again afterwards

Navigation

Before navigation, we must switch on Gyroscope

```
GYR1
```

We can then change the "pitch" of the satellite

```
PIT15
PIT-20
```

As well as the "yaw"

```
YAW-20
YAW10
```

Try these out now, but...
careful with fuel consumption!

Sun Reacquisition

Due to all our previous manual navigation actions...
The satellite can quickly become out-of-alignment:

- Alignment with the Sun (for solar power)
- Alignment with the Earth (for radio control signals)

Luckily satellite provides an "auto pilot" feature:
"Emergency Sun Reacquisition"

ESR

Resets the satellite to the correct orientation
Try running this command now